

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018 - 2019 (Odd Semester)

Innovative Practices Description

UNIT III – SYNCHRONOUS SEQUENTIAL CIRCUITS

Degree, Semester & Branch: III Semester B.E. ECE B

Course Code & Title: EC8392 Digital Electronics

Name of the Faculty member: Mr.R.Deiva Nayagam

Name of the Topic: Counters

Name of the Innovative Practice: Real Time Applications

Description:

A digital circuit which is used for counting pulses is known counter. Counter is the widest application of flip-flops. It is a group of flip-flops with a clock signal applied. Counters are of two types. Asynchronous and Synchronous counters.

I explained some real world applications which are using counter circuits. I show some videos related to applications of counters, which are the essential part of our day to day life.

In everyday life we are using synchronous counter in all digital circuits which have timing in its. Suppose in the morning you will wake up with the help of alarm and the TV and AC and sports watch and digital watches we are using synchronous counters. These counters find their applications in embedded systems, stop watch, real-time clock and processors. After receiving the clock signal they count the desired sequence in order.

I motivated the students to do a mini project with the help of counters. I gave the electronic components and other materials in the laboratory classes for doing the projects with counter applications. Some of the students successfully designed their mini projects like Automatic Room light controller, Count down applications, Player time counter and Automatic visitor counter.

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	3	3	3	3				1		2		2

CO3: The students will be able to analyze and design synchronous sequential circuits.

References:

1. https://nptel.ac.in/content/storage2/nptel_data3/html/mhrd/ict/text/106105185/lec45.pdf
2. [https://en.wikipedia.org/wiki/Counter_\(digital\)](https://en.wikipedia.org/wiki/Counter_(digital))
3. <https://www.geeksforgeeks.org/counters-in-digital-logic/>